

## Q1:JUN 2016 P2

- \*8 (a) (i) State any two assumptions of the kinetic theory of gases.

1. \_\_\_\_\_

2. \_\_\_\_\_

- (ii) Explain the term *internal energy of an ideal gas*.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

[4]

- (b) Deduce, by comparing  $PV = \frac{1}{3}Nm \langle c^2 \rangle$  with  $PV = NkT$ , that the average translational kinetic energy of a molecule is proportional to T.

[2]

- (c) A sealed flask of volume  $100 \text{ cm}^3$  contains argon gas at a pressure of 10 kPa and temperature of  $30^\circ\text{C}$ . Molar mass of argon is 0.018 kg.

Calculate the r.m.s speed of the molecules in the flask.

r.m.s speed = \_\_\_\_\_ [2]

## Q2:JUN 2008 P2

- 9 (a) Define internal energy of a gas.

\_\_\_\_\_  
\_\_\_\_\_ [1]

- (b) (i) State the first law of thermodynamics.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_ [2]

- (ii) The volumes of 1 kg of water and 1 kg of steam at  $100^{\circ}\text{C}$  and  $1.01 \times 10^5 \text{ Pa}$  are  $0.005 \text{ m}^3$  and  $1.500 \text{ m}^3$  respectively. If the specific latent heat of vaporisation of water at  $100^{\circ}\text{C}$  is  $2.26 \times 10^6 \text{ Jkg}^{-1}$ , calculate the change in internal energy of the water molecules if 1 kg of water is changed into steam under these conditions.

change in internal energy = \_\_\_\_\_ [2]

## Q3:NOV 2012 P2

- 8 (a) State **four** assumptions of the kinetic theory of ideal gases.

1. \_\_\_\_\_  
\_\_\_\_\_  
2. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_ [4]

# 6.5 IDEAL GASES P2, P3&P5 QUESTIONS

Compiled by: R. Nuga

- (b) Nitrogen gas enclosed in a cubic container is assumed to behave ideally at 10 °C.

Given that the molar mass of nitrogen is 28 g, determine the r.m.s speed of nitrogen molecules.

r.m.s. speed = \_\_\_\_\_ [2]

Q4:JUN 2017 P2

- 3 (a) State any **two** basic assumptions of the kinetic theory of gases.

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[2]

- (b) The pressure,  $p$ , of an ideal gas of density  $\rho$ , is given by

$$p = \frac{1}{3} \rho \langle c^2 \rangle.$$

- (i) State the meaning of the symbol  $\langle c^2 \rangle$ .

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- (ii) Show that the average kinetic energy of the molecules is proportional to the thermodynamic temperature,  $T$ .

[4]

## Q5:JUN 2019 P2

4 (a) (i) State the

1. *zeroth law of thermodynamics,*

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2. *first law of thermodynamics.*

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[2]

(ii) Calculate, the thermodynamic temperature given that

$$\frac{(pV)_T}{(pV)_{tr}} = 2.230 \text{ for an ideal gas at temperature } T.$$

[2]

## Q6:NOV 2016 P2

8 (a) State the ideal gas equation.

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[1]

6.5 IDEAL GASES P2, P3&P5 QUESTIONS

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- (b) The density of nitrogen at standard temperature and pressure (s.t.p) is  $1.25 \text{ kg m}^{-3}$ . The s.t.p values are  $1.01 \times 10^5 \text{ Pa}$  and  $273 \text{ K}$ .

Calculate the r.m.s speed,  $c_{r.m.s}$ , of nitrogen molecules at  $298 \text{ K}$ .

$c_{r.m.s} =$  \_\_\_\_\_ [3]

- (c) Explain how an increase in internal energy of an ideal gas is related to a rise in temperature of the ideal gas.

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[2]

Q7:JUN 2009 P5

- 4 (a) Give any **one** assumption of the kinetic theory of gases. [1]
- (b) A cubical box contains a carbon dioxide gas molecule. Show that the force the molecule exerts on the wall is proportional to the square of its speed. [3]

(c) Fig. 4.1 shows a graph of pressure against density of a gas at  $25^{\circ}\text{C}$ .

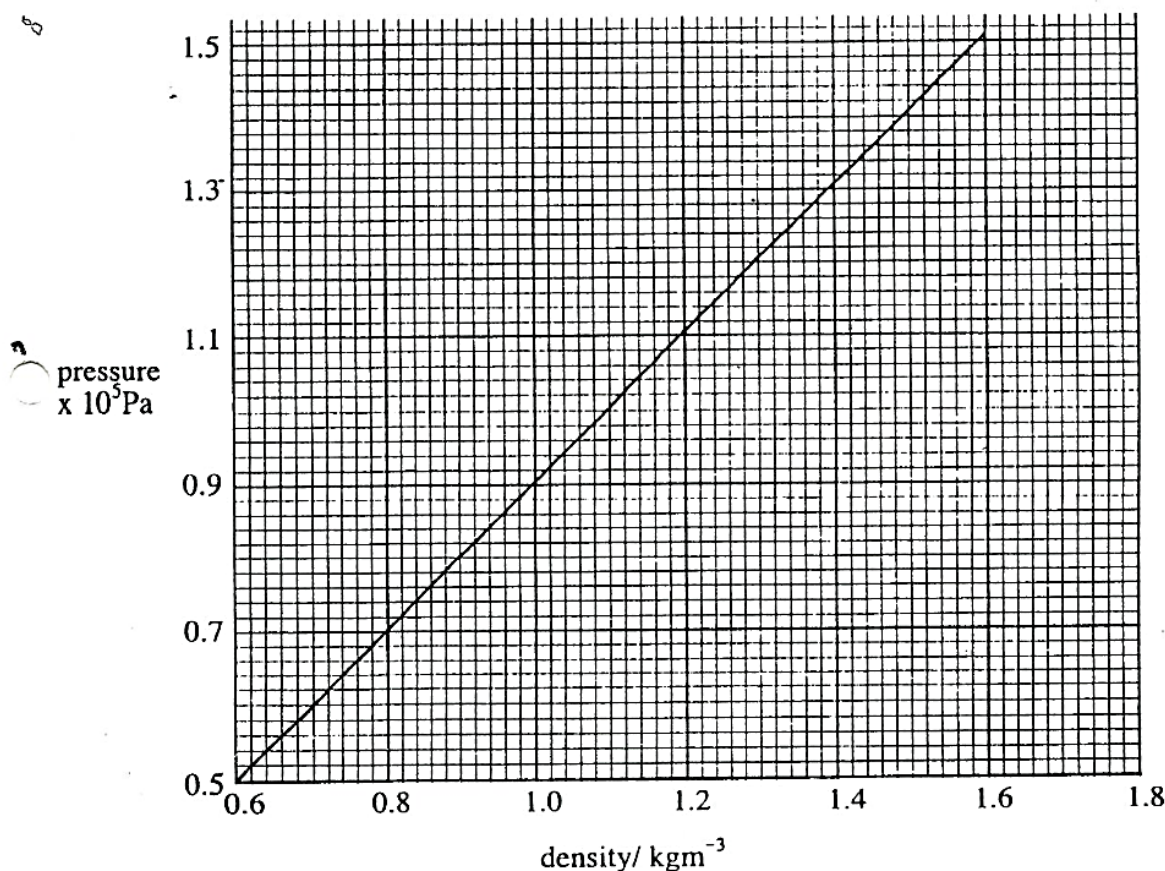


Fig. 4.1

- (i) State and explain whether or not the gas behaves as an ideal gas at this temperature.
- (ii) Use the graph to calculate the root mean square speed.
- (iii) The temperature of the gas is raised to  $55^{\circ}\text{C}$  calculate the pressure when the density is  $1.02 \text{ kg m}^{-3}$ .

[8]

Q8:NOV 2009 P5

- 1 (b) (i) A sealed container of volume  $0.8 \times 10^{-3} \text{ m}^3$  contains a gas at a temperature of  $47^{\circ}\text{C}$  and a pressure of  $1.5 \times 10^6 \text{ Pa}$ .

Calculate

1. the number of molecules of the gas,
  2. the kinetic energy of the molecules when pressure is increased to  $2.5 \times 10^6 \text{ Pa}$ .
- (ii) Give any **two** assumptions made when coming out with the solutions above.

[8]

## Q9:NOV 2010 P5

- 4 (a) State **four** basic assumptions of an ideal gas. [4]
- (b) (i) State what is meant by *internal energy of a system*.  
(ii) Explain why the internal energy of an ideal gas is wholly kinetic. [2]
- (c) When  $150\text{ J}$  was supplied to an ideal gas at  $1000\text{ kPa}$  its volume changed from  $2,5 \times 10^{-4}\text{ m}^3$  to  $5,8 \times 10^{-4}\text{ m}^3$ .  
Determine the  
(i) potential energy of the gas,  
(ii) work done by the gas,  
(iii) change in internal energy and state if this is an increase or decrease. [6]

## Q10:JUN 2014 P5

- 3 (b) (i) State *the first law of thermodynamics*.  
(ii) Explain the term *internal energy of a system*.  
(iii) State how the internal energy of a system can be increased.  
(iv) Fig. 3.1 shows a sample of a gas enclosed in a cylinder by a frictionless piston of surface area  $150\text{ cm}^2$ .

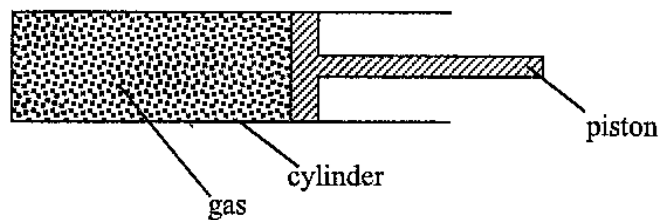


Fig. 3.1

When  $300\text{ J}$  of energy is supplied to the gas, it expands against an atmospheric pressure of  $1 \times 10^5\text{ Pa}$  and pushes out the piston for a  $16\text{ cm}$  distance along the cylinder.

Calculate the change in internal energy.

[6]

## Q11:JUN 2011 P2

- 2 (a) Distinguish between *mean speed* and *mean square speed* of a group of molecules. [2]
- (b) Explain how gases exert pressure on the walls of a container of the gas. [3]
- (c) The density of hydrogen gas at  $0^\circ\text{C}$  and  $1.01 \times 10^5 \text{ Pa}$  pressure is  $9.0 \times 10^{-2} \text{ kg m}^{-3}$ .  
Determine the root-mean-square speed of hydrogen molecules. [4]
- (d) In a reaction chamber, chlorine, hydrogen and oxygen in gaseous state at same temperature are being mixed.  
A leak develops on the chamber and it is noticed that the reaction could not proceed because hydrogen mainly had escaped.  
Explain this observation given the densities at one atmosphere are  $3.2 \text{ kg m}^{-3}$ ,  $1.2 \text{ kg m}^{-3}$ , and  $0.09 \text{ kg m}^{-3}$  for chlorine, oxygen and hydrogen respectively. [3]

## Q12:JUN 2013 P5

- 1 (a) (i) Explain the terms in the ideal gas equation of state.  
(ii) One molecule of oxygen was put in a  $500 \text{ cm}^3$  container.  
Calculate the temperature of the oxygen gas if the molecule is moving at  $590 \text{ m/s}$ .  
(iii) State and explain whether your answer in (ii) is reliable or not. [5]

## Q13:NOV 2014 P5

- 5 (a) (i) Define *pressure*.  
(ii) Show that the pressure exerted by an ideal gas on the wall shown in Fig. 5.1 is given by  $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$ .

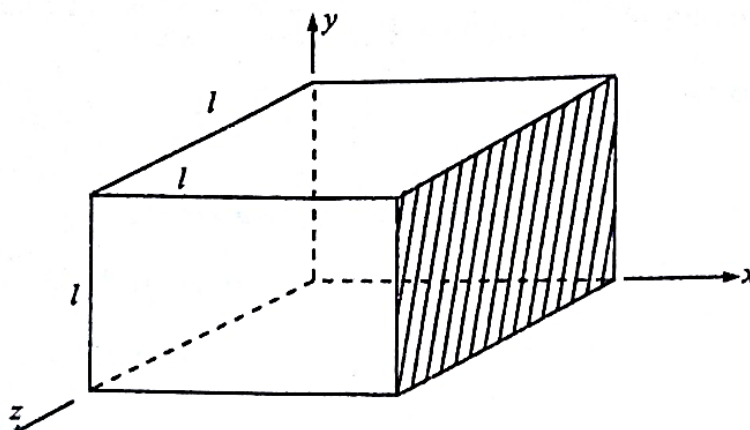




Fig. 5.1

[5]

- (b) Fig. 5.2 shows some oxygen compressed by a mercury column in a long J-tube.

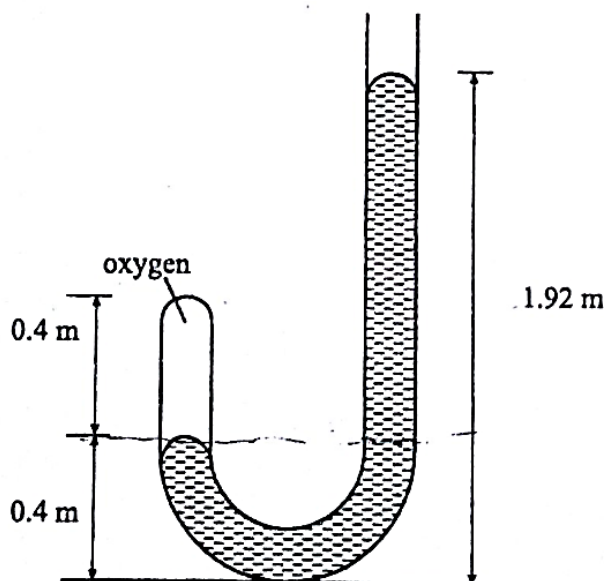


Fig. 5.2

Calculate the pressure of oxygen given that the atmospheric pressure is  $1.01 \times 10^5$  Pa and density of mercury is  $13\,600 \text{ kg/m}^3$ .

[2]

- (c) A car tyre contains a compressed gas at a pressure of  $3.42 \times 10^5$  Pa and temperature of  $3^\circ\text{C}$ .
- Calculate the number of molecules of the gas in the tyre if the volume of the tyre is  $1.5 \text{ m}^3$ .
  - Explain why after travelling a long distance the pressure of the gas increases.

[5]

Q14:JUN 2015 P5

- 3 (a) State the equation of state of an ideal gas identifying the terms used.
- (b) In the derivation of the equation,

[2]

$$p = \frac{1}{3} \rho \langle c^2 \rangle.$$

state the significance of the factor  $\frac{1}{3}$

[2]

- (c) A soccer ball has a constant volume of  $4.5 \text{ l}$  and contains an ideal gas at a pressure of  $2.03 \times 10^5 \text{ Pa}$  at  $20^\circ \text{C}$ . The pressure is to be increased to  $2.60 \times 10^5 \text{ Pa}$  by pumping in more of the gas at  $0.0041$  moles per stroke.

(i) Calculate

- the number of strokes required at a temperature of  $28^\circ \text{C}$ .
- the root-mean square speed of the molecules of the gas at  $28^\circ \text{C}$  if each molecule has a mass of  $14 \text{ u}$ .

(ii) Show that the internal energy of the gas in the ball at  $20^\circ \text{C}$  and a pressure of  $2.03 \times 10^5 \text{ Pa}$  is  $1.37 \text{ kJ}$ .

[8]

Q15: NOV 2015 P5

- 1 (e) (i) Fig.1.1 shows the variation of pressure,  $P$ , with volume,  $V$ , of a gas.

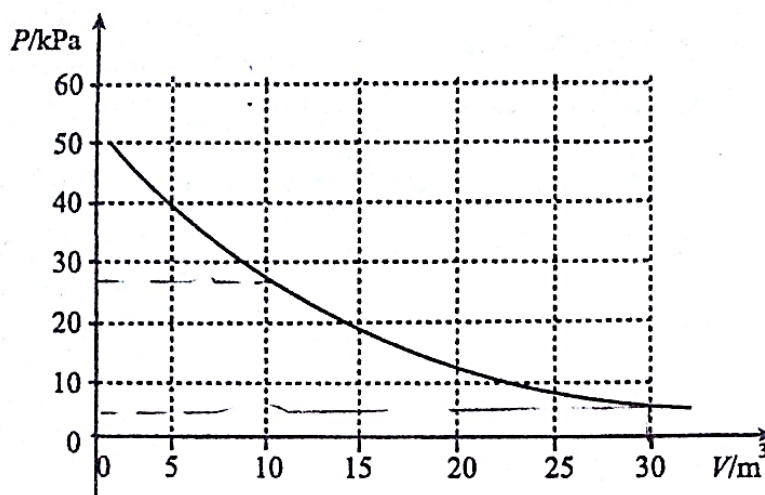


Fig.1.1

Calculate the work done by the gas as the volume increases from  $10 \text{ m}^3$  to  $30 \text{ m}^3$ .

- (ii) Suggest why a hot air balloon expands as it rises.

[3]

# 6.5 IDEAL GASES P2, P3&P5 QUESTIONS

Compiled by: R. Nuga

## Q16:JUN 2017 P5

- 1 (b) (i) Write what each symbol in the equation of state,  $pV = nRT$ , for an ideal gas represents.
- (ii) 1. Calculate the number of moles of hydrogen gas at a temperature of  $18.0^\circ\text{C}$  and pressure  $2.35 \times 10^6 \text{ Pa}$  in a cylinder of volume  $2.40 \times 10^{-3} \text{ m}^3$ .
2. State any assumption you made in the calculation in (b)(ii)1.

[5]

## Q17:NOV 2003 P3

- 5 (d) (i) Using the equation  $pV = \frac{1}{3}Nm\langle c^2 \rangle$  and the ideal gas equation, deduce that the average kinetic energy of a gas molecule is  $\frac{3}{2}kT$ .
- (ii) Hydrogen gas is contained in a cylinder of volume  $1.2 \times 10^{-3} \text{ m}^3$ . The pressure of the gas is  $1.0 \times 10^5 \text{ Pa}$  at a temperature of  $300 \text{ K}$ . Calculate
- 1 the number of hydrogen atoms in the container,
- 2 the total kinetic energy of the hydrogen atoms.

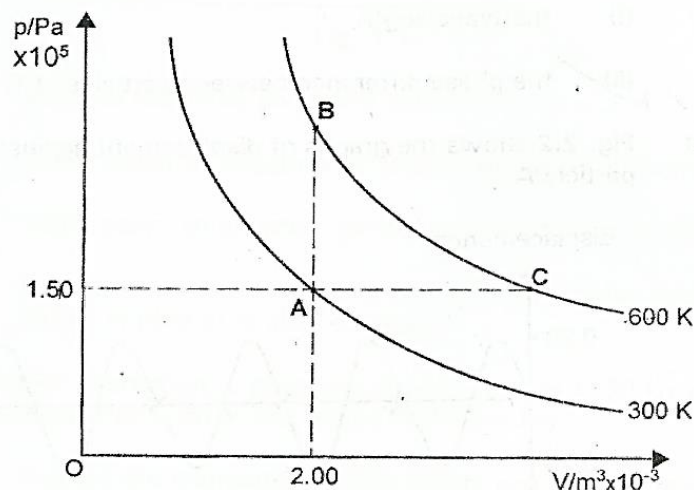
[4]

[5]

## Q18:NOV 2002 P3

3 (a)

Fig. 3.1 shows curves (not drawn to scale) relating pressure  $p$ , and volume  $V$ , for a fixed mass of an ideal monatomic gas at temperature  $300 \text{ K}$  and  $600 \text{ K}$ . The gas is enclosed in a container fitted with a piston which can move with negligible friction.



## 6.5 IDEAL GASES P2, P3&P5 QUESTIONS

Compiled by: R. Nkomo

- (i) State the ideal gas equation for  $n$  moles of the gas, explaining the symbols used.
- (ii) Calculate the number of moles of the gas.
- (iii) Calculate the volume of gas corresponding to point C on the graph. [6]
- (b) The pressure of an ideal gas is given by  $p = \frac{1}{3} \rho \langle c^2 \rangle$  where  $\rho$  is the density of the gas.
- (i) What is meant by  $\langle c^2 \rangle$ ? [1]
- (ii) Using the equation above, derive an expression for the total internal energy of one mole of an ideal monoatomic gas at kelvin temperature  $T$ . [2]
- (iii) Calculate the total internal energy of the gas in the container corresponding to point A on the graph. [8]
- (c) (i) State the first law of thermodynamics.
- (ii) Using the first law of thermodynamics, calculate the heat absorbed during changes represented on the graph from A to B and A to C. [6]

### Q19: NOV 2007 P3

- 6 (a) (i) State any **three** assumptions made in the kinetic theory of gases.
- (ii) Suggest why oxygen enclosed in a gas cylinder at room temperature is assumed to behave ideally. [5]
- (b) (i) The ideal gas equation can be expressed in the form

$$PV = \frac{1}{3} Nm \langle c^2 \rangle$$

State the meaning of each symbol in the equation.

- (ii) A  $0.15 \text{ m}^3$  container has 3.1 moles of oxygen. The root mean square speed of the molecules is 455 m/s. Calculate the pressure exerted on the walls by the gas. Mass of an oxygen molecule is 32u. [6]

## Q20:JUN 2008 P3

5 (a) State **four** assumptions of the kinetic theory of gases. [3]

(b) Derive the equation of total pressure  $P = \frac{1}{3} \frac{Nm \langle c^2 \rangle}{V}$ , where symbols have their usual meaning. [4]

(c) Oxygen gas of mass 2.2 kg is in a gas cylinder. The density of the gas is  $1.28 \text{ kg m}^{-3}$  at a temperature of  $39^\circ\text{C}$ . Given that the molar mass of oxygen is  $32 \text{ g/mol}$  calculate

(i) the number of oxygen molecules in the cylinder,

(ii) the root mean square speed of the molecules,

(iii) the pressure exerted by the gas on the container. [7]

(d) The gas in (c) absorbs heat until the temperature is  $50^\circ\text{C}$ . The *specific heat capacity* of oxygen is  $913 \text{ J kg}^{-1}\text{K}^{-1}$ .

(i) State what is meant by *specific heat capacity*.

(ii) Starting from the first law of thermodynamics, calculate the change in internal energy of the gas. State any assumptions that you make. [6]

6 (a) In an adiabatic change no heat is supplied or extracted.

(i) Write down an expression of the first law of thermodynamics for an adiabatic change in the state of an ideal gas in which the gas does 600 J of work.

(ii) Comment on the temperature of the gas. [5]

## Q21:NOV 2018 P3

4 (d) (i) Explain what is meant by *mean square speed* of an ideal gas.

(ii) The density of a gas at  $25^\circ\text{C}$  and pressure of  $10^5 \text{ Pa}$  is  $1.29 \text{ kg m}^{-3}$ .

Calculate the mean square speed,  $\langle C^2 \rangle$ , of the gas.

- (iii) Determine the temperature of a gas if a gas molecule has a mean kinetic energy of  $5.0 \times 10^{-21}$  J. [6]
- (e) Nitrogen has a melting point of 82 K and a boiling point of 96 K. Distinguish between nitrogen at 73 K and nitrogen at 273 K, in terms of
1. intermolecular spacing,
  2. motion of molecules. [2]

Q22:NOV 2016 P5

- 5 (a) (i) State any three assumptions in the derivation of the equation

$$P = \frac{1}{3} \frac{Nm \langle c^2 \rangle}{V}$$

- (ii) Suggest the effect of the collisions of gas particles being inelastic. [6]
- (b) Fig. 5.1 shows the pressure-volume variation in an engine cylinder from intake of fuel through to release of waste gases.

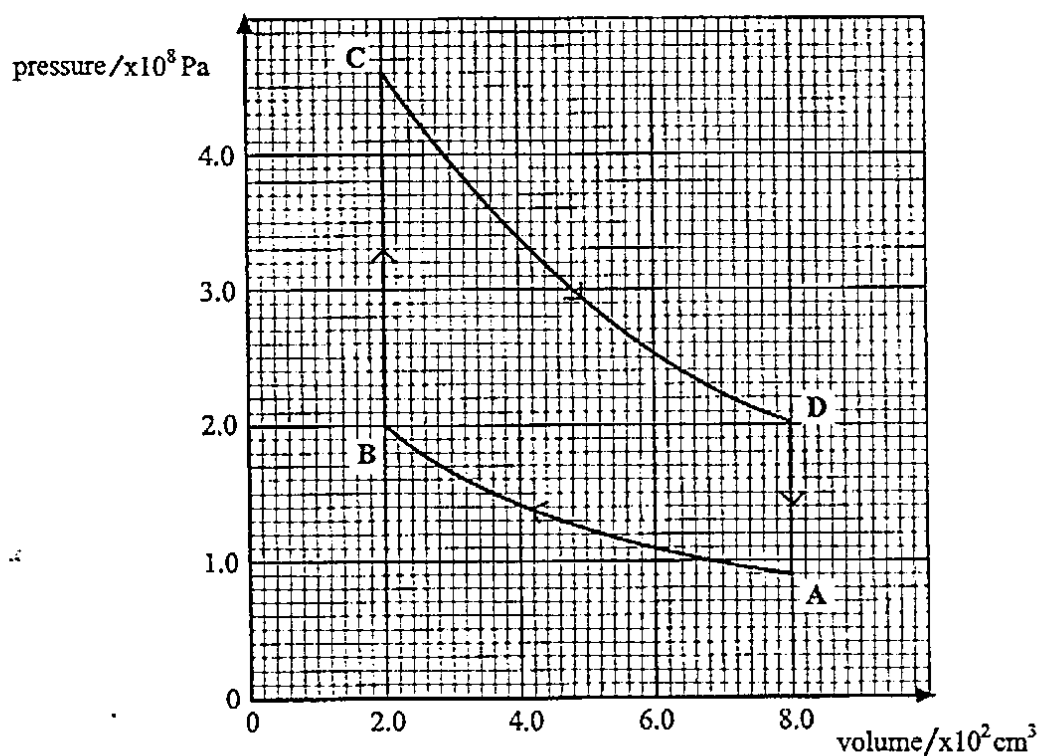


Fig. 5.1

## 6.5 IDEAL GASES P2, P3&P5 QUESTIONS

*Compiled by: R. Nuga*

- (i) Estimate the net work done by the cylinder piston in one cycle.
- (ii) Describe the changes occurring shown by the line BC and DA.
- (iii) Suggest how the efficiency of the system can be improved.

[6]